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COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

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Quiz - Exam

Duration: 70 minutes --- October 23, 2025

Both Problems must be done

Problem 1: (50 points)

Let's design a shell and tube heat exchanger to heat a stream with $m_c C_{p,c} = 5,000 \text{ BTU} / \text{h}^\circ\text{F}$ from 90°F to 210°F flowing in the tubes using as the heat source a stream with $m_h C_{p,h} = 10,000 \text{ BTU} / \text{h}^\circ\text{F}$ entering the shell at 260°F . The heat exchanger will contain tubes with a 0.5 in. outside diameter. In order to maintain a reasonable fluid velocity inside the tubes, $N_t = 30$ tubes per pass will be used. The overall heat transfer coefficient is $U_o = 150 \text{ BTU} / \text{h ft}^2 \text{ }^\circ\text{F}$. For a heat exchanger with one-tube pass and one-shell pass, determine the required tube length. **(40 points)**

Suppose that we need to have shorter tubes, what can we do? **(10 points)**

$$q = \dot{m}_c (p, c) (T_{c,o} - T_{c,i})$$

$$q = 5000 \frac{\text{BTU}}{\text{h}^\circ\text{F}} (210^\circ\text{F} - 90^\circ\text{F})$$

(120)

$$q = 600000 \frac{\text{BTU}}{\text{h}^\circ\text{F}}$$

To solve for $T_{h,out}$

$$q = \dot{m}_h (p, h) (T_{h,in} - T_{h,o})$$

$$600000 \frac{\text{BTU}}{\text{h}^\circ\text{F}} = (260^\circ\text{F} - T_{h,o}) \times (10000 \frac{\text{BTU}}{\text{h}^\circ\text{F}}) \rightarrow T_{h,o} = 200^\circ\text{F}$$

Since now we know both Temperatures, we can use

LMTD for a counterflow $\rightarrow q = UA \Delta T_m$

$$q = UA \left[\frac{\Delta T_2 - \Delta T_1}{\ln[\Delta T_2 / \Delta T_1]} \right] \quad \Delta T_1 = T_{h,i} - T_{c,o}$$
$$\Delta T_2 = T_{h,o} - T_{c,i}$$

$$\Delta T_1 = 260^\circ\text{F} - 210^\circ\text{F} \rightarrow \Delta T_1 = 50^\circ\text{F}$$

$$\Delta T_2 = 200^\circ\text{F} - 90^\circ\text{F} \rightarrow \Delta T_2 = 110^\circ\text{F}$$

$$\text{Using } Q_{\max} \rightarrow q = 600,000 \frac{\text{BTU}}{\text{h}\cdot\text{F}}$$

$$600,000 \frac{\text{BTU}}{\text{h}\cdot\text{F}} = 150 \frac{\text{BTU}}{\text{ft}^2\cdot\text{F}} \cdot A \cdot \left[\frac{110^\circ\text{F} - 50^\circ\text{F}}{\ln[110/50]} \right]$$

Solving for A (area)

$$52.56 \text{ ft}^2 = A$$

The area is $A = \pi P_i$ or $A = \pi D_o$

I assume that $D_i = D_o$ so $D = 0.5 \text{ in}$

$$D = 4.1667 \times 10^{-2} \text{ ft}$$

Since we got 30 tubes

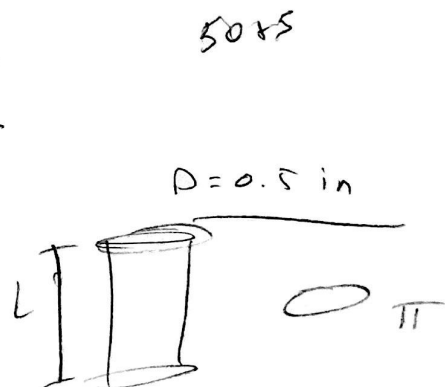
$$\frac{30(\pi)(4.1667 \times 10^{-2} \text{ ft})}{\text{ft}} \rightarrow 3.927 \frac{\text{ft}^2}{\text{ft}}$$

Length can be defined as

$$L = \frac{\text{Area (ft}^2\text{)}}{(\# \text{ of tubes})(\pi)(D) \text{ ft}}$$

$$L = \frac{52.56 \text{ ft}^2}{3.927 \text{ ft}} \rightarrow L = 13.38 \text{ ft}$$

This is going to be the length of each one of the tubes

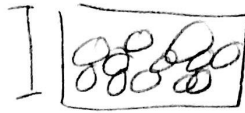


If we suppose that there are shorter tubes, we can increase the number of tubes.

For example, if the "new" length of the tubes is $\frac{1}{2}$ of the original length, then we need to increase the number of tubes by 2 times.

This means that they are going to have the same area split among the tubes.

Or we could also increase the tube diameter to compensate the "lost" in length of the tubes, meaning that if we increase the circumference the area per unit length increases



1L water

Problem 2: (60 points)

Let's consider a case where 1 kg of aluminum spheres with radius $R = 10$ mm initially at 90°C are immersed in 1 L of water initially at 20°C in a tank. The tank is insulated and heat losses are negligible. The water is assumed to be perfectly mixed. Determine the time needed to cool the aluminum spheres to 40°C assuming that the external heat transfer resistance at the sphere-fluid interface is dominant and that $h = 500 \text{ W/m}^2\text{K}$. The density of aluminum is $\rho = 2,700 \text{ kg/m}^3$ and its heat capacity is $C_P = 900 \text{ J/kg K}$. Assume that the water temperature remains constant. (30 points)

If the changing temperature of the water is considered, determine the time needed to cool the aluminum spheres to 40°C . (30 points)

Transient problem

$$\frac{T(t) - T_\infty}{T_c - T_\infty} = \exp\left(-\frac{hA_s}{\rho V_c} t\right) \quad V_c = C_P$$

Solve for t

$$\ln\left(\frac{T(t) - T_\infty}{T_c - T_\infty}\right) = -\frac{hA_s}{\rho V_c} t$$

$$t = \frac{\rho V_c}{-hA_s} \cdot \ln\left(\frac{T(t) - T_\infty}{T_c - T_\infty}\right)$$

$$\text{Sphere volume} = \frac{4}{3} \pi R^3$$

$$V_s = \frac{4}{3} \pi (1 \times 10^{-2} \text{ m})^3 \rightarrow V_s = 4.19 \times 10^{-6} \text{ m}^3$$

$$\text{The mass of an sphere} \rightarrow m_s = \rho(V_s)$$

$$m = 0.0113 \text{ kg}$$

number of sphere $N = \frac{1}{0.0113} = 89 \text{ spheres}$

$$\text{Total surface} \rightarrow A_t = N(4\pi R^2) = 0.111 \text{ m}^2$$

$$t = - \frac{m_s C_{p,s}}{h A_t} \ln \left(\frac{T(t) - T_\infty}{T_i - T_\infty} \right)$$

$$t = - \frac{900 \text{ J/K}}{(\frac{500 \text{ W}}{2.2 \text{ K}})(0.111 \text{ m}^2)} \ln \left(\frac{40 - 20}{90 - 20} \right) \quad 30+5$$

$$t = 20.32 \text{ seconds} \quad \text{Part A}$$

Part B: Since now changing of water is considered

$$m_s C_s T_s + m_w C_w T_w = \text{constant (Not change)}$$

$$C = (900)(90) + (4180)(20) = 164600 \text{ J}$$

$$\text{The eq temp is } T_{eq} = \frac{164600 \text{ J}}{900 + 4180 \text{ J/K}} = 32.40 \text{ }^{\circ}\text{C}$$

$$\alpha = \frac{h A (m_s C_s + m_w C_w)}{m_s C_s + m_w C_w}$$

$$\alpha = \frac{(500/19)(5080)}{900 + 4180} \rightarrow \alpha = 0.0750 \text{ s}^{-1}$$

30

From cooling equation

$$T_s(t) = T_{eq} + (T_{s0} - T_{eq}) e^{-\alpha t}$$

$$46^{\circ}\text{C} = 32.40^{\circ}\text{C} + (90 - 32.40^{\circ}\text{C}) e^{-0.0750 \text{ s}^{-1} t}$$

$$\text{Solving for } t \rightarrow 7.60 = (90 - 32.40) e^{-0.0750 \text{ s}^{-1} t}$$

$$\ln 0.1319 = -0.0750 \text{ s}^{-1} t$$

$$-2.025 = -0.0750 \text{ s}^{-1} t$$

$$t = 27.00 \text{ sec}$$

Part B